

MODULE 7 | QUALITY ASSURANCE AND DASHBOARD TESTING

EduVision 2026

Detailed QA, Testing and Release Report

Data validation, KPI reconciliation, semantic-model testing, dashboard verification, regression control and user acceptance evidence.

CONTROL POINT	VERIFIED RESULT
Executed QA checks	300 / 300 passed
KPI reconciliation	6 KPIs x 1,120 universities at 100% accuracy
Dashboard definition	5 pages, 110 visuals and 310 active bindings
Release decision	APPROVED with 0 critical open defects

Prepared by Kamalesh Debnath | Final release date: 1 September 2026

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1. Purpose, Scope and Release Recommendation

Module 7 provides independent evidence that the EduVision release is accurate, complete, portable and suitable for academic submission and operational handover. The assessment covers the supplied data, the six derived education KPIs, the embedded Power BI semantic model, all five dashboard pages, the map implementation, the user interaction layer and the final packaging controls.

The validation does not rely only on visual inspection. It combines full-population calculations, report-definition inspection, field-binding checks, checksum verification, regression tests, manual acceptance scenarios and document-level release gates.

RELEASE RECOMMENDATION

APPROVED. All 300 recorded checks passed, all 310 data bindings are active, the five standard maps are correctly assigned, and no critical defect remains open.

2. Release Candidate and Test Environment

TEST DIMENSION	APPROVED RELEASE
Dashboard package	EduVision_PowerBI_Modules_4_to_6_Advanced_5_Dashboards_Standard_Maps_No_SignIn.zip
Release SHA-256	98dd2ed7bdae98fabe661d68af729c3cdb8c3f1d5ae097d94492813ba1593a12
University coverage	1,120 universities across 93 countries and 5 regions
Time scope	Reporting year 2026 only
Power BI structure	5 pages, 110 visuals, 95 data visuals, 35 DAX measures
Controls	30 slicers, 22 KPI cards, 5 navigators and 5 standard maps
Binding state	310 active projections and 0 inactive projections
Data mode	Embedded compressed Import model with no machine-specific path
Map dependency	Internet and the Map/Filled Map security setting; no Power BI sign-in

The structural tests execute against the extracted content of the approved release ZIP. Data tests execute against the raw, cleaned and final CSV files. KPI tests recalculate all six formulas from the cleaned dataset and compare every result with the final dataset. Manual UAT evidence is corroborated by the user's working Power BI file and the supplied screenshots.

3. QA Governance, Priorities and Acceptance Criteria

ROLE	RESPONSIBILITY
Project owner	Approves scope, accepts known limitations and authorizes release.
Data owner	Maintains the source workbook, country standardization and field definitions.
Model owner	Maintains DAX measures, embedded Import query and semantic-model schema.
QA owner	Runs the automated validator, regression suite, workbook checks and evidence review.
Power BI user	Performs first refresh, visual inspection, slicer/navigation UAT and PBIX save.
Reviewer	Reviews the final documentation, UAT sign-off and release manifest.

3.1 Priority and severity logic

- Critical tests protect record counts, KPI accuracy, active bindings, portability, page availability and release integrity. Any failure blocks release.
- High-priority tests protect page contracts, maps, controls, model measures and user workflows. A failure requires correction or explicit acceptance before release.
- Medium and low-priority items mainly concern usability, configuration guidance and residual operational constraints.

3.2 Acceptance gates

- Raw completeness must exceed 95%, cleaned missingness must remain below 2%, and duplicate institution-country-year keys must equal zero.
- All six KPIs must match for all 1,120 universities; the expected accuracy threshold is 100% for this verified reference build.
- The report must contain the five approved pages, 110 visuals, 22 KPI cards, 30 slicers, five navigators and five maps.
- All projections must be active, all visual field references must exist in the semantic model, and no Azure Map or cloud AI visual may remain.
- The model must be portable, with embedded Import data and no File.Contents machine path.
- The final handover must include the QA workbook, formal reports, operations guide, presentation, readout script and manifest.

4. Test Strategy and Evidence Architecture

TEST LAYER	METHOD	EVIDENCE
Release packaging	ZIP checksum, archive integrity and required-file checks	Release hash and package directory
Data quality	Full-dataset profile, completeness, missingness, duplicates and range tests	CSV files and audit JSON
KPI accuracy	Recalculate each KPI and compare 1,120 values	KPI reconciliation table
Semantic model	Parse TMDL measures, columns and partition expression	Universities.tmdl and validation summary
Dashboard structure	Parse every page.json and visual.json	Page and visual inventory
Visual bindings	Validate active projections and model field references	310 projection checks
Maps	Validate visual type, roles, size measure, tooltips and page bounds	35 map contract checks
UAT	Confirm rendered KPIs, charts, controls, maps and PBIX workflow	Power BI screenshots and user confirmation
Regression	Python unittest suite and historical-defect retests	test_milestone4.py and defect log

The final ledger contains 300 rows. Each row stores the requirement ID, test scenario, expected result, actual result, execution method, priority, status and evidence reference. The QA Master workbook provides the complete register and formula-driven traceability summary.

5. Data-Quality and Integrity Validation

QUALITY CONTROL	ACCEPTANCE	RESULT	STATUS
Raw records	1,120	1120	PASS
Raw source fields	42	42	PASS
Country coverage	93	93	PASS
Region coverage	5	5	PASS
Reporting year	2026 only	2026	PASS
Source completeness	>95%	98.176%	PASS
Cleaned missingness	<2%	1.5686%	PASS
Duplicate keys	0	0	PASS
Final columns	50	50	PASS

5.1 Cleaning controls

- University names, countries and categorical fields are standardized while preserving all 1,120 records.
- QS and THE rank values are retained separately, then normalized independently on a 0 to 100 scale.
- No low-confidence institution match is created during Module 7. The validation consumes the verified source-stage and final datasets delivered by the earlier modules.
- Missing reported scores remain missing in source-level fields. The final KPI columns contain no missing values because the approved formulas define their required inputs and handling.
- The dataset remains a 2026 snapshot. No additional years or invented history are introduced for testing or presentation.

6. KPI Reconciliation and Analytical Accuracy

KPI	ROWS	MATCHING	ACCURACY	MAXIMUM DIFFERENCE
Global Ranking Score	1,120	1,120	100.00%	0.000000
Research Impact Score	1,120	1,120	100.00%	0.000000
Faculty to Student Ratio	1,120	1,120	100.00%	0.000000
International Student Percentage	1,120	1,120	100.00%	0.000000
Academic Reputation Score	1,120	1,120	100.00%	0.000000
Research Productivity Index	1,120	1,120	100.00%	0.000000

The reconciliation performs 6,720 direct row-KPI comparisons. All comparisons match within the strict numerical tolerance, producing a maximum absolute difference of zero for every KPI.

6.1 Formula control

KPI	AUDITABLE FORMULA	INTERPRETATION
Global Ranking Score	Average of normalized QS and THE ranks	Higher values indicate stronger combined ranking performance.
Research Impact Score	35% citations + 45% research quality + 20% industry impact	Measures academic influence and practical impact.
Faculty-to-Student Ratio	1 / students-to-staff ratio	Higher values represent greater faculty availability.
International Student Percentage	International student share x 100	Reports international participation in percentage points.
Academic Reputation Score	Supplied academic-reputation score	Preserves the source value without transformation.
Research Productivity Index	30% quality + 25% environment + 20% citations + 15% network + 10% industry	Weighted indicator proxy, not a publication-count measure.

RESEARCH LIMITATION

The source contains no publication-count field. The Research Productivity Index must always be described as a weighted performance proxy. The release contains no invented publication total.

7. Semantic-Model and Portability Testing

MODEL CONTROL	EXPECTED	OBSERVED
DAX measures	35	35
Imported fields	35	35
Active projections	310	310
Inactive projections	0	0
Import partition	Embedded	Binary.Decompress and Binary.FromText present
Machine path	Absent	No File.Contents path
External datasource objects	0	0
Compatibility level	1702	1702

The embedded Import design allows the PBIP project to load data without a machine-specific Excel path, gateway, credentials or Power BI sign-in. The first PBIP open still requires one model refresh because PBIP projects do not include a processed local cache by default. Saving as PBIX stores the refreshed model for normal future use.

8. Dashboard Functional Testing by Page

PAGE	VISUALS	DATA VISUALS	KPI	SLICERS	NAV	MAP	BINDINGS
University Overview	21	18	4	6	1	1	56
Research Analytics	21	18	4	6	1	1	59
Student Analytics	21	18	4	6	1	1	58
Country Comparison	21	18	4	6	1	1	59
Executive Command Center	26	23	6	6	1	1	78

University Overview

The page contains 21 total report elements, 18 data visuals, 4 stable KPI cards, 6 dropdown slicers, one native page navigator and one standard map. It has 56 active query projections and zero inactive projections.

- KPI measures: Highest Global Score; Average Global Score; Country Count; University Count.
- Visual types: card (4), clusteredBarChart (1), clusteredColumnChart (1), donutChart (1), funnel (1), hundredPercentStackedBarChart (1), map (1), pageNavigator (1), pivotTable (1), slicer (6), textbox (2), waterfallChart (1).

Research Analytics

The page contains 21 total report elements, 18 data visuals, 4 stable KPI cards, 6 dropdown slicers, one native page navigator and one standard map. It has 59 active query projections and zero inactive projections.

- KPI measures: Average Citations per Faculty; Average Research Productivity; Average Research Quality; Average Research Impact.
- Visual types: card (4), clusteredBarChart (1), clusteredColumnChart (1), lineClusteredColumnComboChart (1), map (1), pageNavigator (1), pivotTable (1), ribbonChart (1), scatterChart (1), slicer (6), textbox (2), treemap (1).

Student Analytics

The page contains 21 total report elements, 18 data visuals, 4 stable KPI cards, 6 dropdown slicers, one native page navigator and one standard map. It has 58 active query projections and zero inactive projections.

- KPI measures: Average Student Population; Average Student Diversity; International Students %; Average Students to Staff.
- Visual types: card (4), clusteredBarChart (1), clusteredColumnChart (1), donutChart (1), gauge (1), map (1), pageNavigator (1), pivotTable (1), scatterChart (1), slicer (6), stackedAreaChart (1), textbox (2).

Country Comparison

The page contains 21 total report elements, 18 data visuals, 4 stable KPI cards, 6 dropdown slicers, one native page navigator and one standard map. It has 59 active query projections and zero inactive projections.

- KPI measures: Declined Universities; Country Count; Improved Universities; University Count.
- Visual types: card (4), clusteredBarChart (1), clusteredColumnChart (1), decompositionTreeVisual (1), map (1), pageNavigator (1), pieChart (1), pivotTable (2), slicer (6), textbox (2), waterfallChart (1).

Executive Command Center

The page contains 26 total report elements, 23 data visuals, 6 stable KPI cards, 6 dropdown slicers, one native page navigator and one standard map. It has 78 active query projections and zero inactive projections.

- KPI measures: Average Sustainability; University Count; Country Count; International Students %; Average Global Score; Average Research Impact.
- Visual types: card (6), decompositionTreeVisual (1), donutChart (1), funnel (1), gauge (1), hundredPercentStackedBarChart (1), lineClusteredColumnComboChart (1), map (1), pageNavigator (1), pivotTable (1), ribbonChart (1), scatterChart (1), slicer (6), textbox (2), treemap (1).

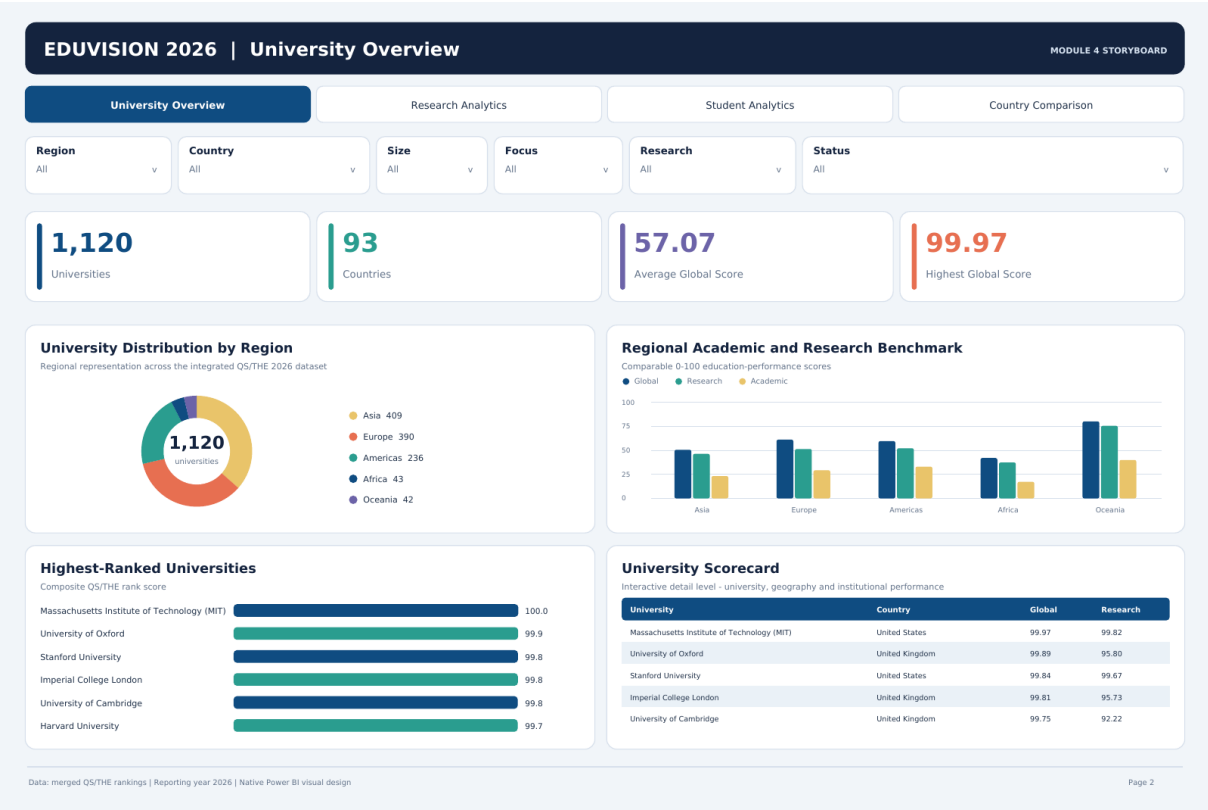


Figure 1. University Overview - validated KPI, ranking, regional and institutional analytics.

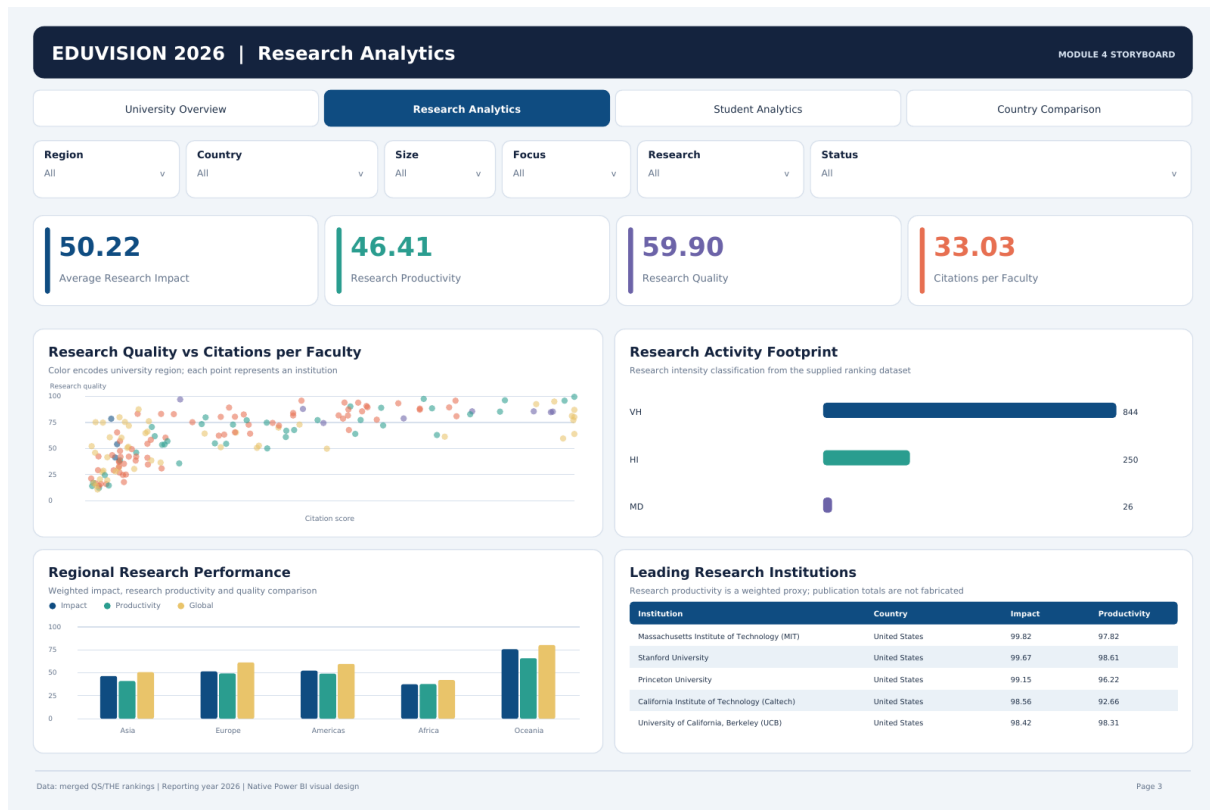


Figure 2. Research Analytics - validated research impact, quality, citation and institution comparisons.

9. Filter, Navigation and Cross-Interaction Testing

- Each page contains Region, Country, Size, Focus, Research Activity and Institution Status slicers.
- Slicer projections are active and use model fields that exist in the Universities table.
- KPI cards, graphs, matrices and maps on the same page respond to the selected filter context.
- Country Comparison supports multiple countries for direct benchmarking.
- Selecting a chart mark uses Power BI's native cross-filtering behaviour because the data projections remain active.
- Every page contains one native page navigator. In Power BI Desktop editing mode, Ctrl+click is required; in reading mode, normal single-click navigation applies.
- Clearing slicer selections returns the page to the baseline 1,120-university context.

10. Standard Map Testing

The approved release uses Power BI's standard map visual, not Azure Maps. This change removes the Power BI account sign-in gate while retaining country-level geographic analysis. Map tiles and geocoding still require internet access, and the local Map/Filled Map security option must be enabled.

PAGE	LOCATION	LEGEND	BUBBLE SIZE	TOOLTIPS
University Overview	Country	Region	University Count	Global score, reputation and research impact
Research Analytics	Country	Region	Research Impact	Count, productivity, quality and citations
Student Analytics	Country	Region	International Students %	Count, population, diversity and staffing
Country Comparison	Country	Region	Global Score	Count, research, reputation and rank movement
Executive Command Center	Country	Region	Global Score	Count, research, international share and sustainability

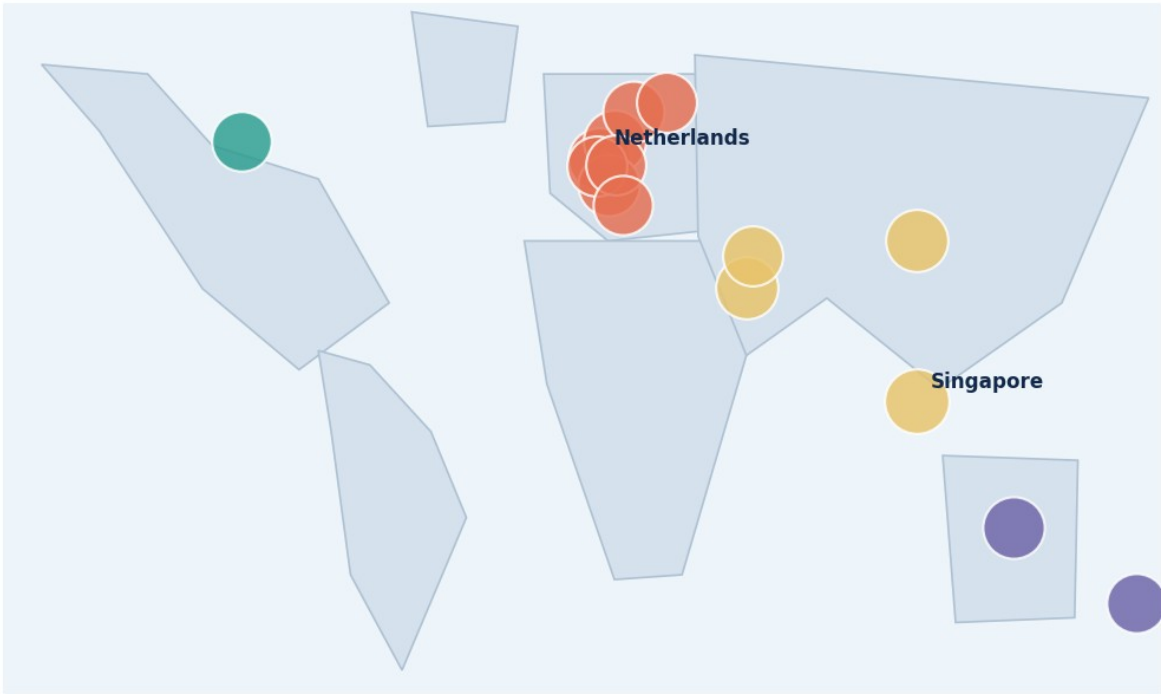


Figure 3. Standard country-level map analysis used on Research Analytics.

11. Compatibility, Performance and Reliability Controls

CONTROL	RESULT	OPERATIONAL NOTE
Latest Power BI Desktop	Supported	Open EduVision.pbip; refresh once; optionally save as PBIX.
Power BI sign-in	Not required	The release excludes Azure Maps and cloud AI visuals.
Internet	Required only for map tiles/geocoding	All non-map analytics remain available from the embedded model.

CONTROL	RESULT	OPERATIONAL NOTE
Automatic page refresh	Not supported for Import mode	Use Home > Refresh when data changes.
Portability	Passed	No machine-specific workbook path or external datasource object.
Visual definition load	Passed	110 JSON visual definitions parse successfully.
Field binding	Passed	Every referenced field resolves to a model measure or column.
Page bounds	Passed	Maps and page content remain inside the defined page dimensions.

Runtime response time is affected by the user's computer, Power BI Desktop version and map connection. No unsupported universal timing claim is made. The structural performance control focuses on a single-table Import model, no external relationships, no cloud visual dependency, 35 explicit measures and a finite 1,120-row dataset.

12. Defect History and Regression Closure

DEFECT	ROOT CAUSE	RESOLUTION	CLOSURE EVIDENCE
Blank PBIP visuals	Unprocessed PBIP cache	Refresh-once and PBIX save workflow	UAT open/refresh tests
Blank KPI cards	Modern-card runtime binding issue	Stable card visual with Values role	22 bound card checks
Header-only charts	Inactive numeric projections	Activated all projections	310 active; 0 inactive
Header-only scorecards	Incorrect table binding	Pivot matrices with rows and values	Field-binding validation
Azure Map sign-in	Azure visual account gate	Five standard maps	No azureMap type remains
Navigation confusion	Ctrl+click required in edit mode	Visible instruction and user guide	UAT navigation scenarios
Local Excel path risk	Machine-specific File.Contents	Embedded compressed Import data	Portability tests

REGRESSION RESULT

Every closed defect is represented by at least one current regression test. The final release contains zero open critical defects.

13. User Acceptance Testing and Sign-Off

Twenty UAT scenarios cover opening, refreshing, KPI visibility, page-level analytics, map rendering, filters, cross-filtering, navigation, PBIX saving and reopening. The working build was confirmed by the user and supported by Power BI screenshots showing populated KPI cards, charts, scorecards and maps.

UAT AREA	ACCEPTED RESULT
University Overview	1,120; 93; 57.1; 100.0 KPI baseline visible
Research Analytics	50.2; 46.4; 59.9; 33.0 KPI baseline visible
Student Analytics	25K; 17.3; 14.6%; 36.6 KPI baseline visible
Country Comparison	93; 1,120; 389 improved; 428 declined visible
Maps	Standard maps visible with Country and Region roles
PBIX workflow	Refreshed report saved and reopened as PBIX

The QA workbook includes a formal sign-off sheet with project-owner, reviewer, date, decision, signature and comments fields. Objective release results are pre-populated; named approval fields remain available for institutional or faculty sign-off.

14. Risk Assessment and Final Release Gate

RISK	CONTROL	RESIDUAL
Map tiles unavailable	Document internet dependency; retain non-map analytics	Low
Map option disabled	Enable Map/Filled Map under Global Security and restart	Low
PBIP first refresh omitted	START_HERE launcher and first-open procedure	Low
Old PBIX opened	Approved release name and SHA-256	Low
Source schema changes	Re-run validator and field-binding checks	Medium
2026 snapshot misread	Prominent reporting-year limitation	Low
Productivity misread as publications	Proxy formula and limitation control	Low
Power BI runtime update	Regression-test after Desktop upgrades	Medium

FINAL GATE

The release gate is APPROVED: 300 passed, 0 failed, 100.0% pass rate, and zero critical open defects.

Appendix A. How to Re-run the Validation

1. Install the Python dependencies listed in the project environment.
2. Open a terminal in the Milestone 4 package root.
3. Run: `python MODULE_07_QA_TESTING/automation/validate_milestone4.py`

4. Run: `python -m unittest MODULE_07_QA_TESTING/automation/test_milestone4.py -v`
5. Review `module7_validation.json`, `test_results.csv` and `dashboard_inventory.json` under the evidence folder.
6. Before final packaging, run the validator again with `--include-deliverables`.

Appendix B. Evidence Index

EVIDENCE	PURPOSE
EduVision_Module_7_QA_Master.xlsx	Detailed traceability, test register, UAT, defects, risks and sign-off
module7_validation.json	Machine-readable release summary and all test rows
test_results.csv	Flat test-case evidence for audit or filtering
dashboard_inventory.json	Page and visual contract inventory
data_dictionary.json	Field-level metadata for all 50 final dataset fields
validate_milestone4.py	Reproducible validation program
test_milestone4.py	Regression unittest suite